**14** **(a)** Show that

$$\arctan\left(\frac{1}{n+1}\right)+\arctan\left(\frac{1}{n^2+n+1}\right)=\arctan\left(\frac{1}{n}\right)\Rightarrow\arctan\left(\frac{1}{2}\right)+\arctan\left(\frac{1}{3}\right)=\arctan 1. \quad \textbf{[1]}$$

**(b)** Use the arctan addition formula in line 23 to show that

$$\arctan\left(\frac{1}{n+1}\right)+\arctan\left(\frac{1}{n^2+n+1}\right)=\arctan\left(\frac{1}{n}\right), \text{ as given in line 39.} \quad \textbf{[4]}$$

**14(a)**

**14(b)**

**(answer space continued on next page)**

**14(b)** **(continued)**